Exercise sheet 1

Classroom task 1.1: Consider the matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

How many ways do you know to characterize invertibility of A? Which one is the most useful to see or prove invertibility of A?

Exercise 1.1:

1. Ask an AI of your choice the following question:

   What is a linear program in standard form?

   Compare the answer to the defintion of **standard form** in the lecture notes.

2. Continue by asking the AI about each of the definitions of **feasible or infeasible vector**, **feasible set**, **feasible or infeasible linear program**, **optimal value**, **bounded or unbounded linear program** and **optimal solution**, e.g.,

   If we consider a linear program in standard form, when is a vector called feasible and when infeasible?

   Do the answers match the definitions in the lecture notes?

For the submission, copy or write down the key parts of the answers of the AI and just tick it, if it is the same definition and if not, comment on the differences. Make sure that, concerning the definitions of the lecture notes, you are able to state a linear program in standard form and explain the bold words to your colleagues in the exercise class **without notes**.

Exercise 1.2: In order to understand what exactly a student is supposed to do to successfully manage all requirements of the course, there are different sources to consult. Since this information is confusing, a student can only spend a total of 30 minutes trying to figure it out in order to not get a lifelong headache. This conflict needs to be solved by an optimization problem. We assign a personal value to each source of information:

1. *Asking a colleague:* value: 4, time: 5 min,
2. *Coming to the first lecture and listen to the general information about the course:* value: 8, time: 9 min,
3. *Read the description in the moodle course:* value: 7, time: 7 min,
   (Click here.)
4. *Read the LV-Karte:* value: 16, time: 15 min,
   (Click here.)

5. *Doing nothing to not risk the headache:* value: 40, time: 30 min.

Since we do not want to rely only on one source, we need to use at least two. Our goal is to maximize the personal value within the time constraints. Formulate the optimization problem (and do not forget to sign up for the exercise groups (Click here.))

Exercise 1.3: The company "FreshShine" produces the following three products:

- Product 1 (Citrus Shower Gel): one unit yields a profit of 3 euros.
- Product 2 (Citrus Shampoo): one unit yields a profit of 5 euros.
- Product 3 (Citrus Body Lotion): one unit yields a profit of 4 euros.

Production requires three machines (A, B, and C), each of which is available only in limited quantity. The production time per unit is as follows:

- Citrus Shower Gel: 1 hour on machine A and 1 hour on machine B.
- Citrus Shampoo: 2 hours on machine A, 1 hour on machine B, and 3 hours on machine C.
- Citrus Body Lotion: 1 hour on machine A, 2 hours on machine B, and 1 hour on machine C.

The available machine capacity per month is:

- Machine A: 170 hours,
- Machine B: 150 hours,
- Machine C: 180 hours.

The company assumes that it can sell all units produced. The goal is to maximize profit. Formulate the mathematical optimization model for this problem.

Submission deadline: 12.10.2026, 12:00